

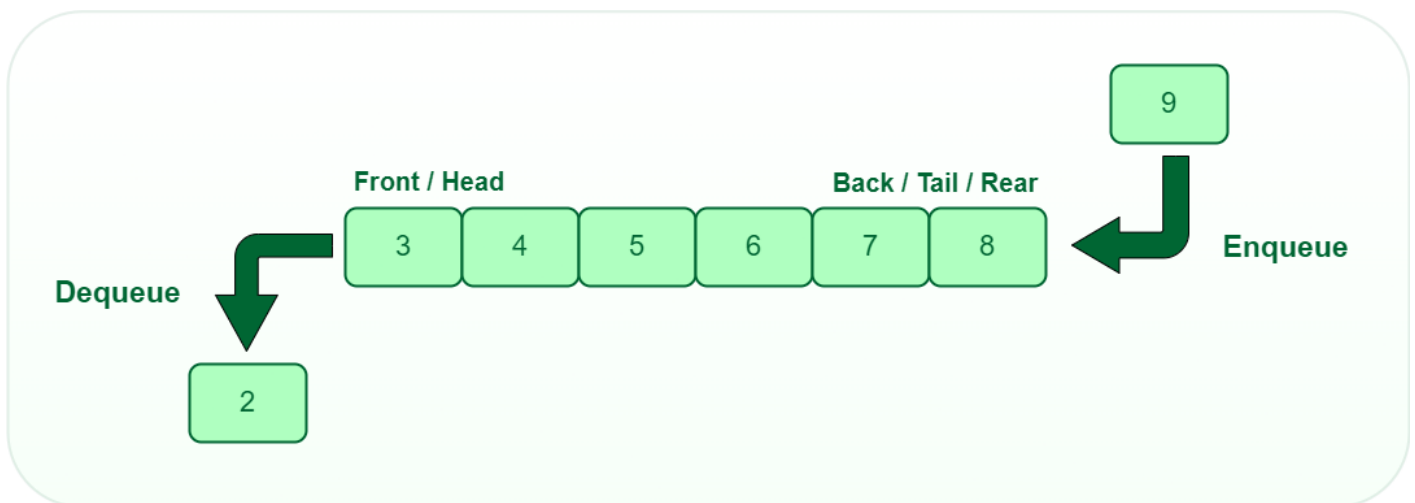
Introduction and Array Implementation of Queue

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Similar to [Stack](#), [Queue](#) is a linear data structure that follows a particular order in which the operations are performed for storing data. The order is First In First Out (**FIFO**). One can imagine a queue as a line of people waiting to receive something in sequential order which starts from the beginning of the line.

- It is an ordered list in which insertions are done at one end which is known as the rear and deletions are done from the other end known as the front.
- A good example of a queue is any queue of consumers for a resource where the consumer that came first is served first.
- The difference between stack and queue is in removing an element. In a stack we remove the item that is most recently added while in a queue, we remove the item that is least recently added.



Queue Data structure

Simple Array implementation Of Queue:

For implementing the queue, we only need to keep track of two variables: **front** and **size**. We can find the rear as $\text{front} + \text{size} - 1$.

- The Enqueue operation is simple, we simply insert at the end of the array. This operation takes $O(1)$ time
- The Dequeue operation is costly as we need remove from the beginning of the array. To remove an item, we need to move all items one position back. Hence this operations takes $O(n)$ time.

- If we do otherwise that insert at the begin and delete from the end, then insert would become costly.

Please refer [Simple Array implementation of Queue](#) for implementation.

Operations	Complexity
Enqueue (insertion)	$O(1)$
Deque (deletion)	$O(n)$
Front (Get front)	$O(1)$
Rear (Get Rear)	$O(1)$
IsFull (Check queue is full or not)	$O(1)$
IsEmpty (Check queue is empty or not)	$O(1)$

Circular Array implementation Of Queue:

We can make all operations in $O(1)$ time using circular array implementation.

- The idea is to treat the array as a circular buffer. We move front and rear using modular arithmetic
- When we insert an item, we increment front using modular arithmetic (which might leave some free space at the beginning of the array).
- When we delete an item, we decrement rear using modular arithmetic.

Please refer [Circular Array implementation Of Queue](#) for details of implementation

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Front (Get front)	O(1)
Rear (Get Rear)	O(1)
IsFull (Check queue is full or not)	O(1)
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